



Power BI Quiz – Advanced DAX

◆ Section A: MCQs (1 mark each)

1. What is the main purpose of the CALCULATE function?

- A. Create columns
 - B. Modify filter context
 - C. Load data
 - D. Create visuals
-

2. Which function is used to apply conditions inside CALCULATE?

- A. SUM
 - B. COUNT
 - C. FILTER
 - D. IF
-

3. Time intelligence functions require:

- A. Only fact table
 - B. Date table
 - C. Customer table
 - D. Product table
-

4. Which function calculates Year-to-Date values?

- A. SUM
- B. FILTER
- C. TOTALYTD
- D. COUNT

5. What does FILTER function return?

- A. A number
 - B. A table**
 - C. A chart
 - D. A column
-

6. CALCULATE function can:

- A. Delete data
 - B. Change filters**
 - C. Create visuals
 - D. Format tables
-

7. Which is correct syntax of CALCULATE?

- A. CALCULATE(value)
 - B. CALCULATE(expression, filters)**
 - C. CALCULATE(table, column)
 - D. CALCULATE(sum, avg)
-

8. What happens if Date table is not connected?

- A. Faster calculation
 - B. Time intelligence won't work**
 - C. Data deleted
 - D. Charts disappear
-

9. FILTER is mainly used for:

- A. Sorting
- B. Conditional filtering**
- C. Aggregation
- D. Grouping

10. YTD calculation depends on:

- A. Region
 - B. Category
 - C. Date column**
 - D. Customer
-
-

◆ **Section B: True/False (1 mark each)**

11. CALCULATE changes filter context.

(**True** / False)

12. FILTER returns a single value.

(True / **False**)

13. TOTALYTD works without a Date table.

(True / **False**)

14. CALCULATE can override existing filters.

(**True** / False)

15. Time intelligence functions require date hierarchy.

(**True** / False)

◆ Section C: Short Answer (2 marks each)

16. What does CALCULATE function do?

Ans : CALCULATE evaluates an expression **by modifying the filter context**. It is the most powerful DAX function used in measures.

17. What is FILTER function?

Ans : FILTER returns a **table of rows** that satisfy a given condition

18. What is Time Intelligence?

Ans : Time Intelligence refers to DAX functions used to perform **date-based calculations** like YTD, MTD, YoY, etc.

19. Why is Date table required?

Ans : A Date table is required to enable **time intelligence functions** and proper date-based filtering.

20. What is YTD calculation?

Ans : YTD (Year-To-Date) calculates **cumulative values from the start of the year till current date**.

◆ Section D: Scenario-Based Questions (3 marks each)

21. Scenario:

You want revenue only for North region.

👉 Which function will you use?

Ans : CALCULATE

22. Scenario:

You want to filter sales where revenue > 2000.

👉 Which function will you use?

Ans : FILTER

23. Scenario:

You want cumulative sales from start of year.

👉 Which function will you use?

Ans : TOTALYTD

24. Scenario:

Your YTD calculation is not working.

👉 What could be the reason?

Ans : No Date table

No relationship with fact table

Incorrect date column

25. Scenario:

You want to change filter inside a measure.

👉 Which function is used?

Ans : CALCULATE

◆ Section E: Practical Thinking (5 marks)

26. Task:

Write DAX formulas for:

1. Revenue for North region
2. High revenue sales (>2000)
3. YTD Sales

Ans :

- Table:Sales
 - Columns:Revenue , Region , Date
-

1. Revenue for North Region

Revenue North =

```
CALCULATE(  
    SUM(Sales[Revenue]),  
    Sales[Region] = "North"  
)
```

2. High Revenue Sales (>2000)

High Revenue =

```
CALCULATE(  
    SUM(Sales[Revenue]),  
    FILTER(Sales, Sales[Revenue] > 2000)  
)
```

3. YTD Sales

YTD Sales =

TOTALYTD(

SUM(Sales[Revenue]),

Sales[Date]

)
